

CHARTING THE FUTURE OF DIGITAL HEALTH SYSTEMS IN THE WHO EUROPEAN REGION

By: Clayton Hamilton

Summary: A key agenda item for European health system decision makers is the role of digital health in reforming and modernising health systems and health service delivery. Only a small number of European Member States have made significant progress in reorienting their health systems to capitalise upon the advantages which digital health and high-quality data can offer. This article seeks to highlight the directions and influences in the digitalization of national health systems, examine the contribution of public health approaches, and offers recommendations on the future of health service design and delivery.

Keywords: Digital Health, Digitization, Health Systems Strengthening, Public Health Approach

Introduction

Digital health* and its role in reforming and modernising health systems and health service delivery has become a key agenda item for European health decision makers. The current level of intensity surrounding digital health and its development is palpable – with several major events on this topic taking place across Europe weekly, each casting a focus on the latest trends and advances in the development of digital services and device offerings for the health sector. Most notably, the recent emphasis on the potential for data-driven technologies, such as Machine Learning and Artificial Intelligence, to disrupt and reform clinical care environments has been a recurring feature of conference headlines

and keynote speeches. Undoubtedly, we are seeing these technologies mature at an unprecedented rate, driven by both substantial private sector investment and a renewed interest by Member States in leveraging the full power of data flowing through their health systems. This, in turn, is rekindling interest in the importance of having timely access to high-quality health data and a re-examination of sources and methods for the capture, codification, storage and exchange of such data.

To many, the unique appeal of digital health lies in its potential to tackle entrenched inefficiencies in health systems and to create a new, tactile paradigm of health care – one in which individuals are empowered through choice; where health-related information is more accessible and actionable, and health services are more transparent and personalised.

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* Defined as a broad umbrella term encompassing eHealth as well as developing areas such as the use of advanced computing sciences (in the fields of, for example, "big data", genomics and artificial intelligence).

Coupled with this vision is a discourse surrounding a shift away from reactive care environments (enabled by digital technologies, big data and genomics-based approaches) to modalities of prediction and disease prevention.

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While a handful of countries in Europe have made significant progress in reorienting their health systems to capitalise upon the advantages which digital health and high-quality data can offer, the reality of the situation across the majority of European countries is a starkly different one. Health systems are still often fraught with piecemeal technology implementations, data interoperability across institutional and regional boundaries is poor, governance and financing for digital health is lacking and health care professionals often feel ill equipped in their use of the available technologies (in addition to feeling overwhelmed by the burden of data entry). Ensuring that these complex, systemic barriers are appropriately addressed by national digitalization programmes requires that the strategic focus remains centred on developing and contextualising the fundamental building blocks of digital health, that investments are aligned to key health policy objectives, and that the trust of health care professionals and the public in their use of digital solutions is well-established.

This article seeks to highlight the directions and influences in the digitalization of national health systems, examine the contribution of public health approaches, and offers recommendations on the future of health service design and delivery.

A public health approach to the digitalization of health systems

As digital health services become more widespread in Europe, a holistic, public health approach to the design of future health systems becomes highly relevant in ensuring that solutions remain safe, accessible and affordable by all segments of the population.

The actions of the World Health Organization (WHO) in developing digital health in countries are anchored in the UN Sustainable Development Goals and in particular, in national targets for Universal Health Coverage (UHC). Mobilising this agenda is taking place through the articulation of WHO's ambitious triple billion target: one billion more people benefiting from UHC; one billion more people better protected from health emergencies; and one billion more people enjoying better health and well-being. Under this framework, WHO seeks to advise countries on how to best position themselves in terms of digitalizing their health systems and data and leveraging the capacity of both, for the delivery of population-wide services and interventions. The notion of digital health as an enabler of public health action is rooted in the understanding that good health and well-being is primarily a function of equity. As such, the delivery of equitable health services is largely dependent upon countries developing well-designed and governed health systems that increase quality and accessibility of health services and facilitate the flow and use of health information at all levels of the system – including transparently to citizens.

The public health approach also acknowledges that significant influences and impacts to our health originate from outside the sphere of influence which care systems provide. Accordingly, the WHO/Europe Health 2020 policy framework

has, for many years, advocated for an intersectoral approach to health that transgresses the silos of government portfolios and funding – calling for shared ownership of strategic goals for health and well-being between multiple stakeholders within and beyond the health domain. Successful efforts for national digital health service development often seek to mirror this intersectoral approach, engaging stakeholders from health and social care, civil society, patient representative groups, Information and Communication Technology (ICT) ministries and the private sector.

Translating goals into action in European Member States

As a vehicle for delivering public health impact in digital health, the WHO/European Regional Office has created the *Digitalization of Health Systems initiative* which seeks to accelerate the safe implementation of digital health in European Member States for the achievement of better health and well-being for all. The initiative includes five overarching dimensions of public health action and support for digital health:

- Designing the future of health service delivery and access;
- Empowering individuals to better manage their own health and well-being through technology;
- Improving the operational efficiency and responsiveness of the health system;
- Enabling the transition to integrated, person-centred models of care and facilitating the move from treatment to prevention;
- Technology and innovation facilitating achievement of key public health initiatives.

Framed under these five dimensions, and in line with the ongoing development of a WHO Global Strategy for Digital Health, the WHO/Europe is developing a roadmap for the digitalization of national health systems that will identify and unpack the key factors contributing to the design of safe, effective and accessible digital health services.

The future of health is digital and personal, but the path remains uncertain and the public health benefits unclear

The rapid expansion and uptake of digital health brings with it many challenging questions to be considered by health decision makers. These include uncertainties regarding approaches to evaluating the safety, efficacy and long-term sustainability of digital solutions in health care; appropriate methods of obtaining and applying consent for the use of health data in different contexts; training of the health workforce; and models of public-private sector engagement that keep the best interests of populations at heart. This is in addition to existential questions being raised as to the influence of technology on the future value base of health systems and their operation.

The reality is that while we are seeing a significant increase in interest in digital health and many efforts to fund and accelerate progress, at the same time, we are observing an increase in solution complexity, new risks appearing, new public concerns emerging, new partnerships, modalities of engagement and methods of value creation. Consequently, governing this landscape requires a new generation of leaders who are adept in the management of continuous change and who can successfully navigate the complex, fluid, multi-faceted, multi-stakeholder environments of digital health.

We are only at the very beginning of seeing the effect of large-scale population-based data analytics being incorporated back into primary health care settings for diagnostics and risk factor identification in individuals. Combined further with omics-based precision medicine approaches that rely “on both biological individuality and population knowledge to provide tailored health care”,¹ we will begin to see an entirely new level of consistency, accuracy and personalisation in predicting, diagnosing and treating disease. This combination represents an incredibly powerful and significant shift in our understanding of what is possible in health care and will correspondingly require restructuring of health systems and services to leverage the massive quantities

of data required for these approaches. We can also expect to see new roles in health care and public health emerge – specifically medical data scientists who configure and apply data analyses to population data sets, and other hybrid medical-technology roles that are capable of actioning such data and developing appropriate care pathways upon this basis.

Governing this landscape requires a new generation of leaders who are adept in the management of continuous change

It is important to note that several, critical questions still remain as to the potential cost implications to public budgets of implementing such approaches and whether or not they will serve to increase or decrease the overall cost of health care and/or overmedicalise health in the long term. As such, we must also consider how to ensure that these approaches do not introduce new inequities and social divisions in our societies, and that benefits are widely and evenly distributed among populations. Further independent research is required to examine the social return on investment of the approaches described above in contrast to, or combination with low-cost public health interventions.

The value of strategic partnerships for digital health

The need to invest further in bold, innovative and far reaching partnerships for digital health cannot be over-emphasised. The momentum of big political interventions and agreements

must be exploited if we are to accelerate progress uniformly in digitalizing European health systems. This includes leveraging partnerships between state actors; patients and representative groups; with young people; with donors and development agencies; with academics and researchers; with developer communities and with the private sector (who are the leaders of innovation). Some examples of the work of partners within the European digital health context include the work done by the European Commission,² through its Digital Single Market Strategy, eHealth Network, and Digital Service Infrastructure approach, which has developed significant national capacity for digital health and cross-border data exchange between EU Member States, the Global Digital Health Partnership³ which represents an independent “collaboration of governments and territories, government agencies and the World Health Organization, formed to support the effective implementation of digital health services” and the OECD⁴ through its agenda for next generation of health reforms which provides recommendations for “Adapting health systems to new technologies and innovation.”

Creating synergies for digital health can only occur when we unite around shared goals for health and well-being. Sustainable Development Goal 17 provides unique momentum to accelerate partnerships, including for digital health, and action within this context should be made to identify trends and best-of-breed approaches, develop evidence on the effectiveness of digital interventions and to define new methods of measuring the performance of digital health systems.

Preparing for the future of digital health systems

To adequately prepare for the future of digital health, Member States are encouraged to embed a public health approach into their national digitalization efforts to ensure that “no one is left behind” in the impending transformation of health services and that social and cultural factors underpinning transformative change in health are taken into consideration in addition to governance, technical, organisational,

financial and process-based requirements. Digitalization is important, but not at any cost. Safety and ethics in digital health are key and there needs to be a clear fundamental caveat of “do no harm” applied to all forms of technology adoption in health.

In designing the future of digital health systems, action by Member States is needed to:

- Develop more concrete approaches to enhancing the digital capabilities of the health workforce, coupled with an open dialogue on inclusive gender approaches and changing roles and responsibilities of health care professionals.
- Build upon the significant achievements already made in the area of standards adoption and interoperability for digital health. This includes development of guidelines on data quality and consent, and clear policy for the use of public health data by private sector entities.
- Focus on the development of broad-scale programmes for digital health literacy.
- Place particular attention on digitalization initiatives deployed in primary health care settings, ensuring

that solutions do not place undue burden on health care professionals and guarantee continuity of care as individuals move through the health system.

“Member States are encouraged to embed a public health approach into their national digitalization efforts”

The development of future health systems in Europe will undoubtedly be shaped in large part by digital technologies. The challenge for digital health and public health communities will be to work together to ensure that the benefits of digitalization are shared equitably and

that services are designed to include and protect the most vulnerable groups in society.

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Spain: Health system review

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The underlying principles and goals of the Spanish national health system continue to focus on universality, free access, equity and fairness of financing. The evolution of performance measures over the last decade shows the resilience of the health system to macroeconomic conditions, although some structural reforms may be required to improve chronic-care management and the reallocation of resources to high-value interventions.

Life expectancy in Spain is the highest in the EU. Inequalities in self-reported health have also declined in the last decade, although long-standing disability and chronic conditions are

increasing due to an ageing population. After decreasing in 2009–2015, public health care spending is on the rise with public sources accounting for over 71.1% of total health financing. Yet private spending, mainly related to out-of-pocket payments, has increased over time, and it is now above the EU average.



Primary care remains the core element of the health system. Public health efforts over the last decade have focused on increasing health system coordination and providing guidance on addressing chronic conditions and lifestyle factors such as obesity. Health system-specific measures to maintain the sustainability of the Spanish health system were implemented in the last decade, with no short-term impact on health outcomes.

Structural measures, however, are needed to improve resource allocation and technical efficiency as well as patients' participation in decisions on their care.